

Module 3 - Requirements Determination

ANALYSIS

TASK CHECKLIST

- ☐ Use requirements elicitation techniques (interview, JAD session, questionnaire, document analysis, or observation).
- ☐ Apply requirements analysis strategies as needed to discover underlying requirements.
- ☐ Develop requirements definition.
- ☐ Develop use cases.
- ☐ Develop data flow diagrams.
- ☐ Develop entity relationship model.
- ☐ Normalize entity relationship model.

Lesson 1: What is a requirement?

A Requirement

- A requirement is **something the system must do**, or a **characteristic** it should have.
- Examples (for a sales system):

The system should ...

1. be accessible to Web users. Non-functional (Operational)
2. include the company standard logo and color scheme. Functional (Information)
3. restrict access to profitability information. Non-functional (Security)
4. include actual and budgeted cost information. Functional (Information)
5. provide management reports. non-functional (Reporting)
6. include sales information that is updated at least daily. Functional (Information)
7. have 2-second maximum response time for predefined queries and 10-minute maximum response time for ad hoc queries. non-functional (Performance)
8. include information from all company subsidiaries. Functional (Information)
9. print subsidiary reports in the primary language of the subsidiary. Non-functional (Cultural)
10. provide monthly rankings of salesperson performance. Non-functional (Reporting)

Functional requirements represent a need for **information** the system should provide, or a **process** it should enable the users to accomplish.

Non-functional Requirements

Nonfunctional Requirement	Description	Examples
<u>Operational</u>	The physical and technical environments in which the system will operate	• The system will run on Android mobile devices. • The system should be able to integrate with the existing inventory system. • The system should be compatible with any Web browser.
<u>Performance</u>	The speed, capacity, and reliability of the system	• Any interaction between the user and the system should not exceed 2 seconds. • The system downloads new status parameters within 5 minutes of a change. • The system should be available for use 24 hours per day, 365 days per year. • The system supports 300 simultaneous users from 9–11 a.m.; 150 simultaneous users at all other times.
<u>Security</u>	Who has authorized access to the system under what circumstances	• Only direct managers can see staff personnel records. • Technicians can see only their own work assignments. • The system includes all available safeguards from viruses, worms, Trojan horses, etc.
<u>Cultural and Political</u>	Cultural and political factors and legal requirements that affect the system	• The system should be able to distinguish between US currency and currency from other nations. • Company policy is to buy computers only from Dell. • Country managers are permitted to authorize custom user interfaces within their units. • Personal information is protected in compliance with the Data Protection Act.

Source: The Atlantic Systems Guild, www.systemsguild.com

FIGURE 3-2 Nonfunctional requirements.

Categories of Requirements

- Business Requirements
- User Requirements
- Functional requirements
- Non-functional requirements
- System requirements

❏ Reading - What Can Happen If You Ignore Nonfunctional Requirements

❏ Minicase - The College's Career Center

Lesson 2: How to discover requirements?

Variety of techniques

- **Analyst as detective**
- The best analysts will thoroughly **search** for requirements using a variety of **techniques**:
 - Interviews
 - Questionnaires
 - Observation
 - JAD Sessions
 - Document Analysis

Selecting the appropriate technique

- Each technique has **strength and weaknesses**.
- Most projects benefit from a **combination** of techniques
- A useful strategy:
 - Get the **big picture**
 - Analyse existing documents and observe actual work to understand **the domain**, vocabulary and the as-is system
 - More interviews

	Interviews	Joint Application Design	Questionnaires	Document Analysis	Observation
Type of information	As-is, improvements, to-be	As-is, improvements, to-be	As-is, improvements	As-is	As-is
Depth of information	High	High	Medium	Low	Low
Breadth of information	Low	Medium	High	High	Low
Integration of information	Low	High	Low	Low	Low
User involvement	Medium	High	Low	Low	Low
Cost	Medium	Low–Medium	Low	Low	Low–Medium

FIGURE 3-12 Comparison of requirements elicitation techniques.

Interviews

- They are used the most
- They can provide a **deep level of information** that uncovers not only the **facts and opinions**, but also an understanding of *why* those facts and opinions exist.
- It is important to use the interviews **effectively** (why?):
 - Don't ask questions about information that you can find from other (indirect) sources
 - Prepare the interview!
 - Prepare the interviewee!

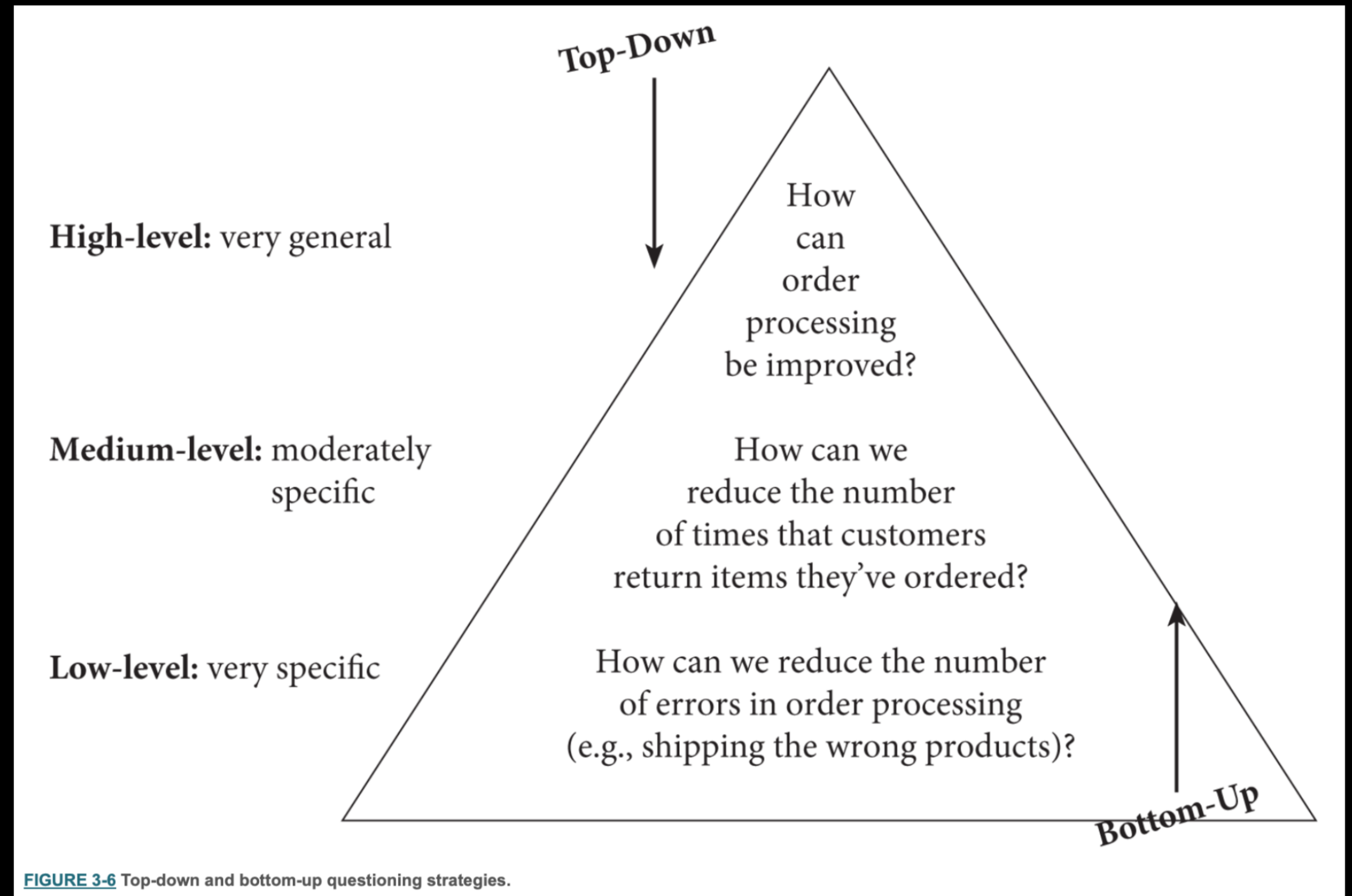
Structured vs unstructured interviews

- **At the beginning**, the as-is process is unclear to the analyst. So she starts with **unstructured interviews**.
 - More *open-ended questions*
 - *Probe* for more information “on the fly”
- As the project progresses, the analyst needs **more specific information**, so she conducts more **structured interviews**.
 - More *close-ended questions*

Preparing for the interview

No matter what kind of interview is being conducted, interview questions must be organized into a **logical sequence** so that the interview flows well:

- **top-down**
- **bottom-up**



Interviewee: Paul H., Human Resources Director

Interview purpose/goals: Understand Paul's requirements for the new performance measures and reporting of measures to be used to monitor the staff of consultants.

Interview strategy: Top-down design. Begin with general discussion of his need for performance measures. Ask open-ended questions about what he is currently missing from the performance measures he has. Explore ideas for new metrics he thinks would help him. Finish with closed-ended questions to confirm understanding of specific metric ideas. Discuss any other areas of concern such as confidentiality issues.

Behavioral issues to expect (if any are known): Expect Paul to be supportive and enthusiastic. He has been asking for these improvements for quite a while.

Interview topics and outline:

General introduction, discuss purpose and goals of meeting

Understand his current approach to understanding the performance of consulting staff

- Current metrics (if any)
- How are these metrics collected and reported (if at all)?
- How are these measures inadequate?

Ideas for improved performance measures

- Develop list of items he thinks would be useful
- Understand the underlying data source of each item—need to know exactly what each one means and how the relevant data is or could be collected

How would he like to receive these metrics?

- Regular reports? If so, how often?
- Dashboard?
- Queries?

Other concerns:

- Are metrics confidential?
 - If yes, identify which ones are confidential and who is allowed to view them.
 - If no, how widely shared will these metrics be?

FIGURE 3-7 Sample interview preparation form.

❏ Reading - Designing Interview Questions

Important: prepare the interviewee as well!

- ❑ How do you think you can do so?

Conducting the interview

Goal: build a **trusting relationship** with the interviewee.

Key: to be professional and unbiased, **independent seeker** of information.

- Do not be afraid to ask “**dumb questions**”. The only thing worse than appearing dumb is to *be* dumb by not understanding something.
- Periodically **summarize the key points**.
- **Separate facts from opinions**
 - Differences between fact and opinion can point out key areas for improvement
- Give interviewee a chance to ask questions or provide information he thinks is important.
- Do not prematurely promise.

PRACTICAL TIP 3-1



Developing Interpersonal Skills

Interpersonal skills are those that enable you to develop rapport with others, and they are very important for interviewing. They help you to communicate with others effectively. Some people develop good interpersonal skills at an early age; they simply seem to know how to communicate and interact with others. Other people are less “lucky” and need to work hard to develop their skills.

Interpersonal skills, like most skills, can be learned. Here are some tips:

- **Don't worry, be happy.** Happy people radiate confidence and project their feelings on others. Try interviewing someone while smiling and then interviewing someone else while frowning and see what happens!
- **Pay attention.** Pay attention to what the other person is saying (which is harder than you might think). See how many times you catch yourself with your mind on something other than the conversation at hand.
- **Summarize key points.** At the end of each major theme or idea that someone explains, you should repeat the key points back to the speaker (e.g., “Let me make sure I understand. The key issues are ...”). This demonstrates that you consider the information important—and also forces you to pay attention. (You cannot repeat what you did not hear.)
- **Be succinct.** When you speak, be succinct. The goal in interviewing (and in much of life) is to learn, not to impress. The more you speak, the less time you give to others.
- **Be honest.** Answer all questions truthfully, and if you do not know the answer, say so.
- **Watch body language (yours and theirs).** The way a person sits or stands conveys much information. In general, a person who is interested in what you are saying sits or leans forward, makes eye contact, and often touches his or her face. A person leaning away from you or with an arm over the back of a chair is disinterested. Crossed arms indicate defensiveness or uncertainty, while “steepling” (sitting with hands raised in front of the body with fingertips touching) indicates a feeling of superiority.

Prepare an interview
report within 48 hours
of the interview.

INTERVIEW NOTES APPROVED BY: LINDA ESTEY

Person Interviewed: Linda Estey,

Director, Human Resources

Interviewer: Barbara Wixom

Purpose of Interview:

- Understand reports produced for human resources (HR) by the current system.
- Determine information requirements for future system.

Summary of Interview:

- Sample reports of all current HR reports are attached to this report. The information that is not used and missing information are noted on the reports.
- Two biggest problems with the current system are:
 1. The data are too old. (The HR department needs information within 2 days of month end; currently information is provided to them after a 3-week delay.)
 2. The data are of poor quality. (Often, reports must be reconciled with the HR departmental database.)
- The most common data errors found in the current system include incorrect job-level information and missing salary information.

Open Items:

- Get current employee roster report from Mary Skudrna (extension 4355).
- Verify calculations used to determine vacation time with Mary Skudrna.
- Schedule interview with Jim Wack (extension 2337) regarding the reasons for data quality problems.

Detailed Notes: See attached transcript.

FIGURE 3-8 Interview report.

Joint Application Development (JAD)

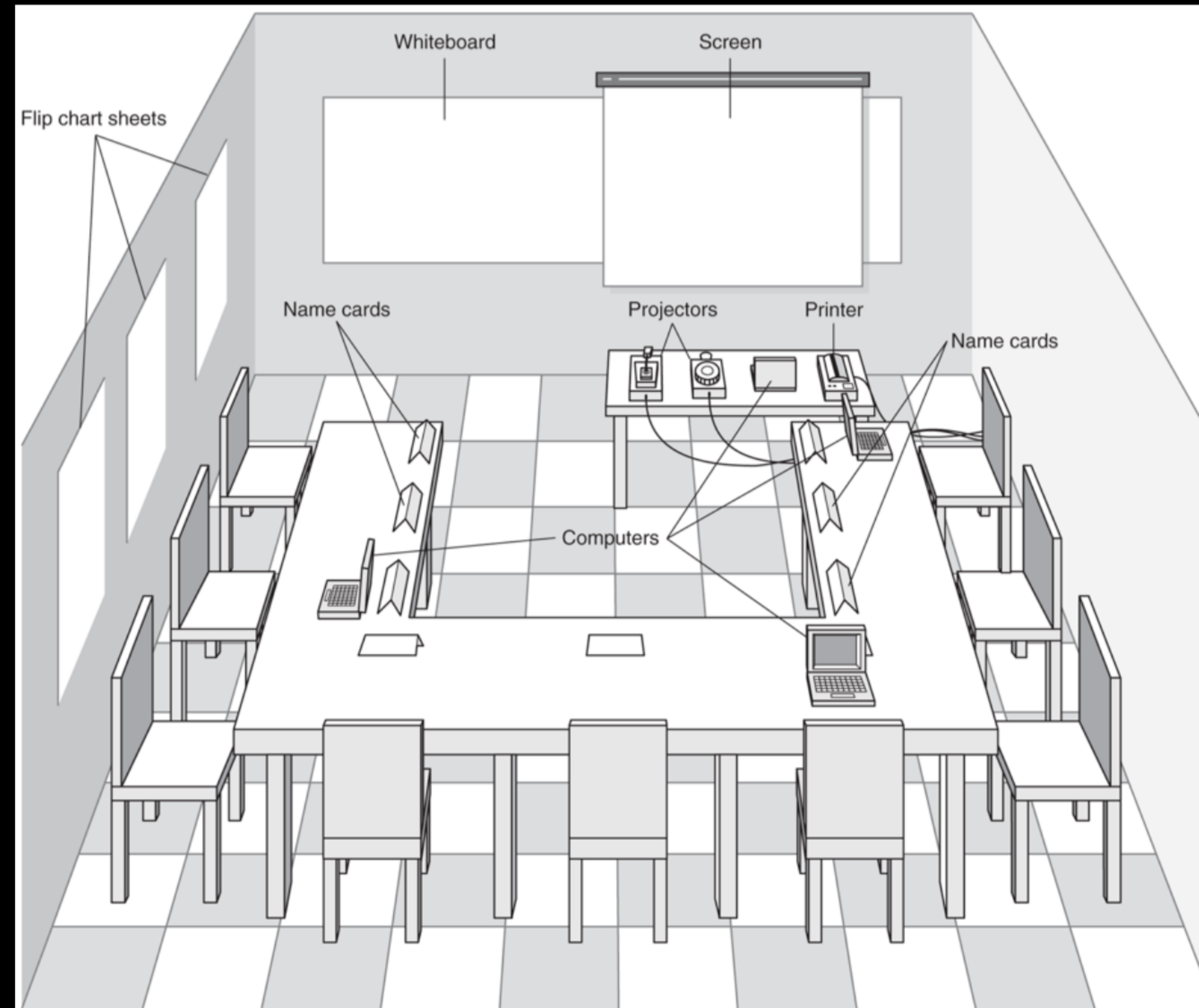


FIGURE 3-9 Joint application development meeting room.

[View in Book](#)

❏ Research JAD

Questionnaires

- Begin with nonthreatening and interesting questions.
- Group items into logically coherent sections.
- Do not put important items at the very end of the questionnaire.
- Do not crowd a page with too many items.
- Avoid abbreviations.
- Avoid biased or suggestive items or terms.
- Number questions to avoid confusion.
- Pretest the questionnaire to identify confusing questions.
- Provide anonymity to respondents.

FIGURE 3-10 Good questionnaire design.

Document Analysis

- To understand the **as-is system**
 - Technical documentation
 - Paper reports
 - Memorandums
 - Policy manuals
 - User training manuals
 - Organization charts
 - Forms
 - ...

Formal vs Informal (Real) system

- Documents represent the formal system.
- Often the real/informal system differs from the formal one.
- So Review both blank / completed forms.

The diagram illustrates a document analysis of a 'CENTRAL VETERINARY CLINIC Patient Information Card'. The card contains the following fields and entries:

- Name: ~~Buffy~~ Pat Smith
- Pet's Name: Buffy Collie 7/6/17 Male
- Address: 100 Central Court, Apartment 10
Toronto, Ontario K7L 3N6
- Phone Number: 416- 555-3400 Cell: 416-567-1369
- Do you have insurance: yes
- Insurance Company: Pet's Mutual
- Policy Number: KA-5493243

Four callout boxes highlight discrepancies between the formal system (the card) and the informal system (the actual data):

- Top Left:** The customer made a mistake. This should be labeled **Owner's Name** to prevent confusion. (Points to the 'Name' field where 'Buffy' is crossed out and 'Pat Smith' is written.)
- Top Right:** The staff had to add additional information about the type of animal and the animal's date of birth and gender. This information should be added to the new form in the to-be system. (Points to the handwritten 'Collie 7/6/17' and 'Male' in the 'Pet's Name' field.)
- Bottom Left:** The customer did not include area code in the phone number. This should be made more clear. (Points to the handwritten '416-' in the 'Phone Number' field.)
- Bottom Right:** The form does not allow for this fact. It should be added to the to-be system. (Points to the handwritten 'Cell: 416-567-1369' in the 'Phone Number' field.)

FIGURE 3-11 Performing a document analysis.

[View in Book](#)

Observation

- Watching processes being performed is a powerful tool to gain insight into the **as-is system**.
- A good way to **check the validity of the info** gathered from other sources: interviews, questionnaires.
- However, know that people behave differently when they are watched, so what you observe might not be exactly the real way people work.

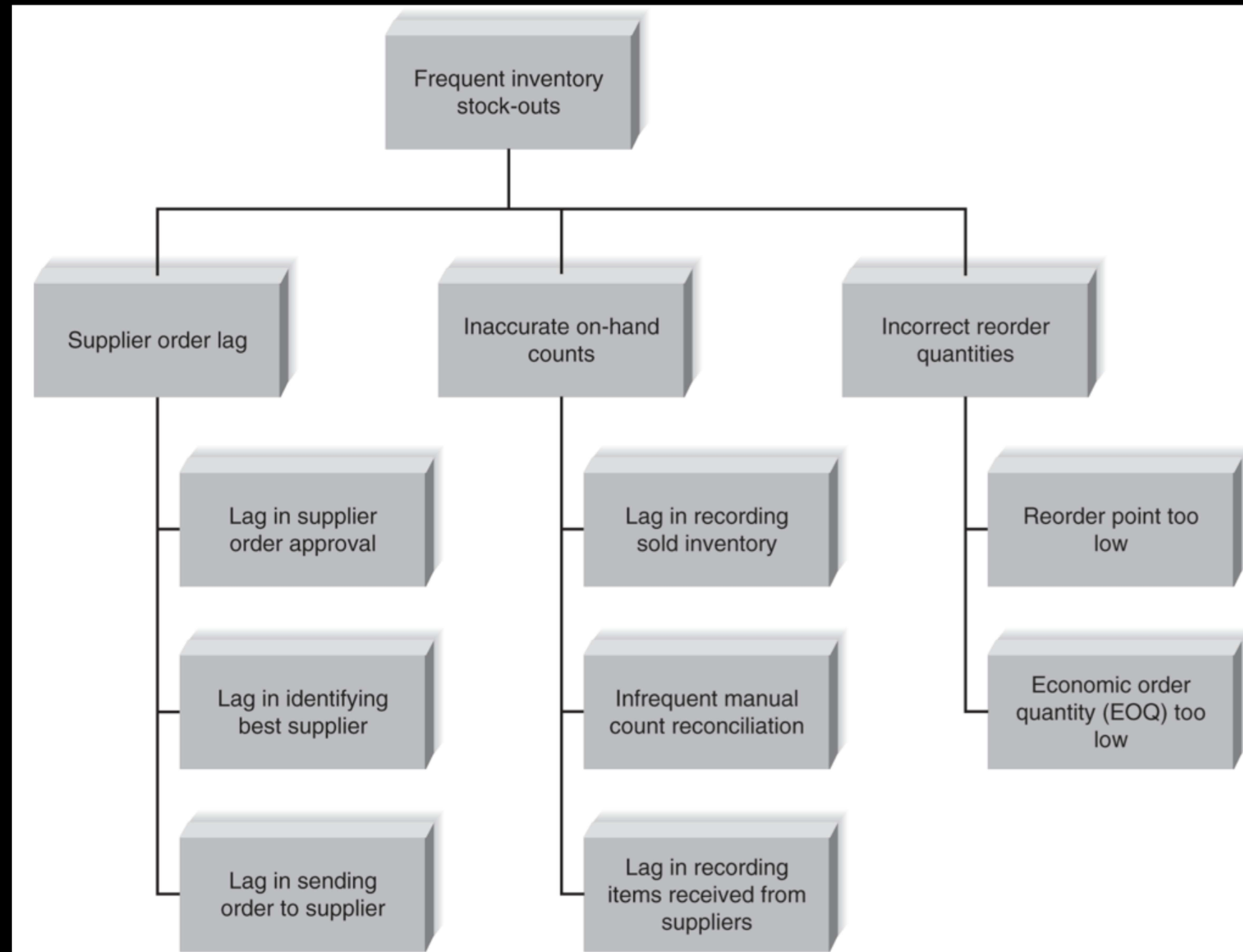
Requirements Analysis Strategies

Problem Analysis

- **Ask stakeholders** to identify problems with the as-is systems and to describe how to solve them in the to-be system.
- Most straightforward and common strategy
- Most changes offered tend to solve problems rather than capitalize on opportunities.
 - Effectively improves a system's efficiency or ease of use. However, it often provides only minor improvements in business value.

Root Cause Analysis

- Users (and most people in general) tend to jump quickly to solutions without fully considering the nature of the problem. Sometimes the solutions are appropriate, but many times they address a **symptom** of the problem, not the true problem or **root cause** itself.
- Root cause analysis focuses on the **problem first** rather than solutions.
- **Key:** Always challenge the obvious and dig into the problem deeply enough that the true underlying cause(s) is revealed.



Apparent
Problem

Possible
Symptoms

Possible
Root Causes

FIGURE 3-13 Root cause analysis for inventory stock outs.

[View in Book](#)

Duration Analysis

- A detailed examination of the amount of time it takes to perform each process in the current as-is system.
 - Begin by determining the total amount
 - Then time each of the individual steps (or subprocesses)
- A significant difference between the two, indicates that the process is badly **fragmented**:
 - Many different people must perform different activities before the process is complete.
- Fragmented processes are prime candidates for process **integration** or **parallelization**
 - Integration → fewer people work on the input
 - Parallelization → all the individual steps are performed at the same time.

Activity-Based Costing

- Similar to duration analysis, but examines the costs of processes rather than the time taken.
- Identifies and focuses on the costliest processes.

Informal Benchmarking

- How do other organizations perform a business process?
- Learn from it!
- Commonly used for “customer-facing” business processes.

Outcome Analysis

- The goal is to understand fundamental outcomes that provide value to customers.
- Encourage stakeholders to pretend they are customers
 - What do we enable our customers to do?
 - What *could* we enable them to do?

Technology Analysis

- Develop a list of important and interesting technologies:
 - How would each technology apply to our business processes?
 - How could our business benefit from each technology?

Activity Elimination

- A “force-fit” exercise:
 - Consider a business process
 - Eliminate each and every activity (in turn)
 - How the function could operate without it?
 - What effects are likely to occur?

Comparing Analysis Strategies

- | | | |
|---------------------------|---|--|
| 1. Problem Analysis | } | Best for projects with a narrow focus where we want to increase efficiency |
| 2. Root Cause Analysis | | |
| 3. Duration Analysis | } | Best for finding the most broken pieces |
| 4. Activity-Based Costing | | |
| 5. Informal Benchmarking | } | Helps participants “ think outside the box ” |
| 6. Outcome Analysis | | |
| 7. Technology Analysis | | |
| 8. Activity Elimination | | |